

Richard Medina Rodríguez

Curriculum Vitae – May 2026

CEREMADE, Université Paris Dauphine - PSL
Place du Maréchal de Lattre de Tassigny, 75016 PARIS
✉ richard.medina-rodriguez@dauphine.psl.eu
🌐 richard-medina.github.io
✂ arXiv 📖 Google Scholar 🆔 Orcid 🔗 LinkedIn

Research Interests

- **Analysis of partial differential equations** (kinetic equations in non-equilibrium regimes, boundary effects, fluid–particle systems).
- **Stochastic differential equations** (propagation of chaos, mean-field limits).
- **Numerical methods for kinetic equations and multiscale systems.**
- **Mathematical modeling motivated by applied problems and interactions with industrial settings.**

Publications

Accepted/Published

- 2026 **Evans, J., Medina, R.**, *Existence and Stability of Non-Equilibrium Steady States of a Weakly Non-Linear Kinetic Fokker-Planck Equation in a Domain*, J Stat Phys 193, 47 [doi](#).
- We work on a nonlinear kinetic Fokker-Planck equation with non-equilibrium forcing via BGK thermostats and non-isothermal Maxwell boundary conditions. We prove well-posedness and long-time convergence to non-equilibrium steady states, extending previous results on the torus.

Preprints

- 2026 **R. Medina**, The Boltzmann equation with non-isothermal Maxwell boundary conditions, URL <https://arxiv.org/abs/2604.13572>.
- We study the Boltzmann equation in non-isothermal bounded domains. We construct non-equilibrium steady states under small boundary temperature fluctuations and prove their exponential asymptotic stability.
- 2025 **R. Medina**, The Boltzmann equation on smooth and cylindrical domains with Maxwell boundary conditions, URL <https://arxiv.org/abs/2510.13260>. Submitted to *Kinetic and Related Models*.
- We study the Boltzmann equation near equilibrium in bounded domains, including smooth and cylindrical geometries, with Maxwell boundary conditions. We prove well-posedness and exponential convergence to equilibrium in a perturbative regime.
- 2024 **K. Carrapatoso, P. Gabriel, R. Medina, and S. Mischler**, Constructive Krein-Rutman result for Kinetic Fokker-Planck equations in a domain, URL <https://arxiv.org/abs/2407.10530>. Submitted to *Analysis & PDE*.
- We study kinetic Fokker-Planck equations in non-isothermal bounded domains. We establish ultracontractivity estimates and use them to prove well-posedness and exponential convergence to equilibrium.

Long-term Research Stays

- Oct. – Nov. 2024 **Warwick Mathematics Institute**, University of Warwick, United Kingdom.
Work in collaboration with Josephine Evans on the effects of interior and boundary thermostats on the long-time behavior of non-linear kinetic Fokker-Planck equations.

Education and Academic Positions

- 2025 – 2026 **Teaching and Research Fellow (A.T.E.R.)**, Université Paris Dauphine - PSL, France.
Research on kinetic theory; teaching assistant for mathematics courses (real analysis and numerical methods).
- 2022 – 2025 **PhD in Applied Mathematics**, Université Paris Dauphine - PSL, France.
Thesis title: Hypocoercivity and geometric confinement.
PhD Advisors: Kleber Carrapatoso & Stéphane Mischler.
Defense date: 18 December 2025.
Thesis reviewers: Véronique BAGLAND & José A. CAÑIZO.
Thesis examiners: Émeric BOUIN, Arnaud GUILLIN & Frédéric HÉRAU (president of the jury).
- 2022 **Master Internship**, CEREMADE, Université Paris Dauphine - PSL, France.
Advisors: Kleber Carrapatoso & Stéphane Mischler.
Research Project: Extending hypocoercivity techniques for linearized kinetic equations to non-smooth bounded domains.
- 2021 – 2022 **Master of Science in Applied and Theoretical Mathematics (M2 MATH)**, Université Paris Dauphine - PSL, France.
With Honors (Mention Bien).
- 2020 – 2021 **Assistant Professor**, *Department of Mathematics*, Universidad de La Habana, Cuba.
Research on functional analysis and PDEs; teaching assistant for mathematics courses (real and complex analysis).
- 2016 – 2020 **Bachelor of Science in Mathematics**, Universidad de La Habana, Cuba.
Summa Cum Laude.
Bachelor's Final Project: Random Schrödinger operators in a 2D scheme.
Advisors: Rita Roldán Inguanzo & Laure Dumaz.

Invited Talks

Invited Talks at Conferences and Workshops

- Jun. 2024 ***The Boltzmann equation in a cylinder near the hydrodynamic limit***, presented at the conference *Analysis of PDEs in Mathematical Physics*, University of Bath, United Kingdom.
- Feb. 2023 ***Weighted L^∞ estimates for the Boltzmann equation***, presented at the *MAFRAN Days*, King's College, United Kingdom.
- Jun. 2019 ***Fixed points in non-expansive maps***, presented at the *International Congress COMPUTAT 2019*, Universidad de La Habana, Cuba.
- Mar. 2019 ***Extrapolation Methods for Approximating a Fairness Functional***, presented at the *International Workshop on Operations Research (IWOR)*, Universidad de La Habana, Cuba.
- Mar. 2018 ***Restriction-based interpolation with cubic A-splines***, presented at the *International Conference on Operations Research (ICOR)*, Universidad de La Habana, Cuba.

Invited Talks at Seminars

- Feb. 2026 ***The Boltzmann equation with non-isothermal Maxwell boundary conditions***, presented at the *SPIKE Seminar*, Institut Henri Poincaré (IHP), France.
- Jan. 2025 ***Brief introduction to the Boltzmann equation***, presented at the *Young Researchers Seminar*, Université Paris Dauphine - PSL, France.
- Nov. 2024 ***Constructive Krein-Rutman result for kinetic Fokker-Planck equations in a domain (joint work with P. Gabriel, K. Carrapatoso and S. Mischler)***, presented at the *Junior Analysis Seminar*, Imperial College London, United Kingdom.
- Oct. 2024 ***The Boltzmann equation on C^1 and cylindrical domains near the hydrodynamic limit***, presented at the *Partial Differential Equations and their Applications Seminar*, University of Warwick, United Kingdom.

- Nov. 2023 ***Introduction to the kinetic Fokker-Planck equation and its long-time behavior***, presented at the Young Researchers Seminar, Université Paris Dauphine - PSL, France.
- May 2023 ***Hypocoercivity estimates for some linearized kinetic operators***, presented at the Young Researchers Seminar, Université de Lille, France.
- Mar. 2023 ***Hypocoercivity estimates for some linearized kinetic operators***, presented at the Young Researchers Seminar, Université Paris Dauphine - PSL, France.

Poster Presentations

- Jul. 2024 ***Constructive Krein-Rutman result for kinetic Fokker-Planck equations in a domain (joint work with P. Gabriel, K. Carrapatoso and S. Mischler)***, presented at the summer school *Collective behavior and Pattern formation*, CIRM, France.

Participation on Conferences, Workshops and Schools

- Jul. 2024 ***Summer school on Collective Behavior and Pattern Formation***, CIRM, France.
- Jun. 2024 ***Research school on Frontiers in Interacting Particle Systems Aggregation-Diffusion Equations & Collective Behavior***, CIRM, France.
- Nov. 2022 ***Research School on Kinetic Theory***, CIRM, France.
- Jul. 2022 ***Conference on When Kinetic Theory Meets Fluid Mechanics***, FIM - Institute for Mathematical Research, ETH Zürich, Switzerland.
- Jun. 2019 ***CIMPA Research Summer School on Mathematical Models in Biology and Related Applications of Partial Differential Equations***, Universidad de La Habana, Cuba.
- Mar. 2019 ***III Winter School in Computer Science on Selected Topics in Probability and Statistics***, Universidad de La Habana, Cuba.

Awards, Fellowships & Grants

- 2022 ***MathInParis2020 COFUND Fellowship*** — PhD grant awarded by the Fondation de Sciences Mathématiques de Paris (FSMP).
- 2021 ***PSL PhD Track Grant 2021*** — Master's Excellence Scholarship awarded by the Université Paris Sciences et Lettres.
- 2020 ***Prize for Scientific Merit*** awarded by the Universidad de La Habana for outstanding scientific work during the Bachelor's studies.

Organization of Seminars and Conferences

- Jun. 2024 Co-organizer of the ***Young Researchers Days*** conference, held at *Domaine de la Tour*, Saint-Pierre-Canivet, France. Three-day conference for PhD students to present and discuss their research. Supported by the Dauphine Doctoral School.
- 2023 – 2024 Co-organizer of the ***Young Researchers Seminar*** at the Université Paris Dauphine - PSL.

Teaching Experience

- Spring 2026 ***Algebra 4 & Numerical methods***, problem-solving sessions for second-year Mathematics students, Université Paris Dauphine - PSL, France.
- Fall 2025 ***Analysis 1***, lectures and problem-solving sessions for first-year Mathematics students, Université Paris Dauphine - PSL, France.
- May 2025 ***Functional Analysis Summer School***, problem-solving sessions for graduate Mathematics students, Universidad de La Habana, Cuba.
- Spring 2024 & 2025 ***Analysis 2***, problem-solving sessions for first-year Mathematics students, Université Paris Dauphine - PSL, France.

- Fall 2023 **Mathematical Methods**, *lectures and problem-solving sessions for first-year Economics students*, Université Paris Dauphine - PSL, France.
- Spring 2021 **Complex Analysis**, *problem-solving sessions for third-year Mathematics students*, Universidad de La Habana, Cuba.
- Fall 2020 **Introduction to Mathematics**, *lectures and problem-solving sessions for first-year Mathematics students*, Universidad de La Habana, Cuba.

Editorial Service

Referee for international journals (*Communications in Mathematical Physics*).

Computer Skills

Programming Languages	PYTHON (advanced), C# (intermediate), HTML (intermediate).
Scientific software	WOLFRAM MATHEMATICA (advanced), MATLAB (intermediate).
Typesetting & visualization	L ^A T _E X (advanced), TikZ/Beamer (advanced).

Languages

Spanish	Native speaker.
English	Bilingual proficiency, TOEIC C1 (June 2022).
French	Bilingual proficiency.

Referees

Prof. Stéphane Mischler

*CEREMADE, Université Paris Dauphine - PSL;
& Institut Universitaire de France (IUF).*

✉ mischler@ceremade.dauphine.fr

Prof. Kleber Carrapatoso

*Centre de Mathématiques Laurent Schwartz (CMLS),
École Polytechnique.*

✉ kleber.carrapatoso@polytechnique.edu

Prof. Josephine Evans

*Warwick Mathematics Institute,
University of Warwick.*

✉ Josephine.Evans@warwick.ac.uk

Prof. Pierre Gabriel

*Institut Denis Poisson,
Université de Tours.*

✉ pierre.gabriel@univ-tours.fr